Fisheries Research 131–133 (2012) 1–8

Contents lists available at SciVerse ScienceDirect

Fisheries Research

journal homepage: www.elsevier.com/locate/fishres

Statistical distribution of age readings of known-age sablefish (*Anoplopoma fimbria*)

Dana H. Hanselman[a,*], William G. Clark[b], Jonathan Heifetz[a], Delsa M. Anderl[c]

[a] *Auke Bay Laboratories, Alaska Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, 17109 Pt. Lena Loop Rd, Juneau, AK 99801, United States*

[b] *International Pacific Halibut Commission, Seattle, WA 98199-1287, United States*

[c] *Resource Ecology and Fisheries Management, Alaska Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, 7600 Sand Point Way NE, Seattle, WA 98115-0070, United States*

ARTICLE INFO

Article history:
Received 11 May 2012
Received in revised form 5 July 2012
Accepted 6 July 2012

Keywords:
Known-age fish
Age reading errors
Geometric distribution
Misclassification matrix
Sablefish
Stock assessment

ABSTRACT

A mark-recapture experiment provided a collection of 172 known-age sablefish from Alaska waters. Otoliths from each fish were read by three readers. The readings have a positive bias among young fish and a negative bias among older fish. Among otoliths of the same age, some tended to give consistently high or low annulus counts, so that the variance of age readings at each true age was about half due to variance among otoliths and half due to variance among replicate readings of individual otoliths. The statistical distribution of age reading errors is well described by an asymmetrical two-sided geometric distribution with age-varying parameters. For comparison, the error distribution was estimated with naive methods that do not use the known ages and that assume the readings are unbiased. These estimated distributions do not match the actual error distributions very well, but they do a surprisingly good job of predicting the distribution of age readings from a stock assessment model's internal estimate of a true age composition. They also produce estimates of recruitment and biomass close to those obtained with the actual error distributions when used in the present sablefish stock assessment.

1. Introduction

Age-based stock assessments rely on estimates of the age composition of commercial and survey catches that are obtained by sampling the catches and determining the ages of the fish in the samples. Often the ages are determined by age readers who count the annuli (annual rings) on otoliths, which are small ear bones extracted from the head of the fish. Sablefish (*Anoplopoma fimbria*) is a long-lived species (up to 84 years in the Gulf of Alaska) that sustains a fishery with an ex-vessel value near U.S. $100 million in Alaska (Hanselman et al., 2011). Therefore, accurate age determination is paramount for stock assessment. For sablefish in particular, the annuli are often difficult to recognize and distinguish from false checks, which are patterns that resemble annuli but are not (Pearson and Shaw, 2004). Age readers strive to be consistent in applying the prescribed criteria for identifying annuli, but age readings of the same otolith can nonetheless vary from reader to reader and even from reading to reading by the same reader. Moreover, all of the age readings of a given otolith can be high or low if the otolith has a number of false checks that look like annuli, or a number of annuli that look like false checks. So a single reading of a single otolith is a random variable: it has some variance and may be biased. In production age reading, each otolith is usually read only once. Replicate readings of some fraction of the otoliths are routinely done for quality control, and these readings provide estimates of the variance among readings but not the bias.

Reading errors in age composition data can be accounted for in a stock assessment by applying an age misclassification matrix to internal model estimates of true age compositions to predict the observed (reader-assigned) age compositions of the samples (e.g. Clark and Hare, 2006; Candy et al., 2012). Specifically, let $b = 1, \ldots, B$ denote true ages and $a = 1, \ldots, A$ denote observed (assigned) ages. The misclassification matrix $\mathbf{M}$ is a matrix with B rows and A columns where each row is the probability distribution of assigned ages $\langle a \rangle$ for a given true age b, so each row sums to one ($\sum_{a=1}^{A} p(a|b) = 1$). In other words, each element of the matrix is $\mathbf{M}[b,a] = p(a|b)$, the probability of assigning age a when the true age is b. If for example the stock assessment model calculates that the true age composition of the catch is $C = \langle p_b \rangle$, a row vector of

* Corresponding author. Tel.: +1 907 789 6626.
E-mail addresses: dana.hanselman@noaa.gov (D.H. Hanselman), wgclark@u.washington.edu (W.G. Clark), jon.heifetz@noaa.gov (J. Heifetz), delsa.anderl@noaa.gov (D.M. Anderl).

0165-7836/$ – see front matter © 2012 Published by Elsevier B.V.
http://dx.doi.org/10.1016/j.fishres.2012.07.004

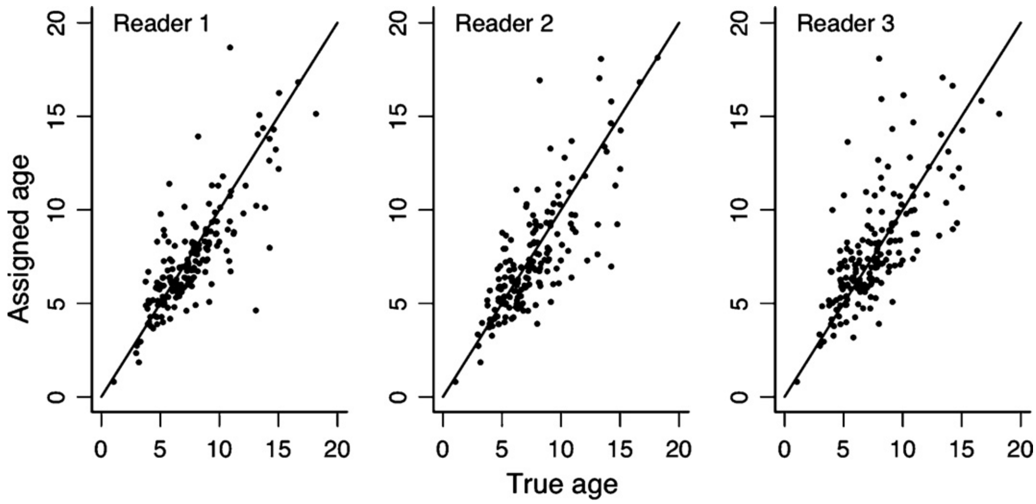

Fig. 1. Plot of ages assigned by each of three readers against true ages. The points were jittered horizontally and vertically to avoid overplotting.

proportions at each true age, then the row vector of expected proportions at each observed (assigned) age $\tilde{C} = \langle p_b \rangle$ can be calculated as $\tilde{C} = C \cdot \mathbf{M}$. In this way the model calculates true age compositions internally and uses the misclassification matrix to predict the observed age compositions.

It is difficult to estimate age reading error because the true ages of the otoliths in the sample are almost always unknown (Conn and Diefenbach, 2007), so the data cannot be simply grouped by true age to estimate the distribution of age readings. Richards et al. (1992) and Punt et al. (2008) estimated the distributions by fitting parametric models to a set of replicate readings of the same otoliths. Clark (2004) developed a nonparametric method applicable when a large number of paired readings are available. Candy et al. (2012) used a continuation ratio method and readability estimates which incorporated asymmetric aging error. However, these approaches assume that unbiased age readings are available. There is no way to detect bias in age readings without knowledge of the true age of the fish.

Existing studies on known-age fish are usually from stocking programs in both freshwater (e.g. McBride et al., 2005) and marine waters (e.g., Cardinale et al., 2004; McDougall, 2004). Many of the species with known-age data are relatively short-lived. Beamish and McFarlane (2000) marked about 30,000 adult sablefish in Canada with oxytetracycline and concluded that sablefish are a difficult species to age and that the age of older fish was often underestimated. Alaska sablefish are almost unique in that we have age readings from a fairly large sample of wild known-age fish now spanning a large age range. These fish were marked early in life (ages 0–2) when the distributions of length at age do not overlap and age at release is therefore known. About 34,000 were marked and released between 1985 and 2005. Heifetz et al. (1999) give a full account of the experiment and estimates of age reading error based on the 49 recoveries available at that time. As result of their findings, age reading criteria were modified. In our study we report on an additional 123 otoliths recovered since the original analysis and read according to the same criteria for a total of 172 otoliths.

The objective of our paper is to update and expand the analysis in Heifetz et al. (1999). We report the statistical features of the raw data and identify a parametric distribution that fits the observed errors very well. We also compare the estimates of age reading error based on known ages with the estimates that would be made in the absence of known ages, and we report on how various ways of treating age reading error affect the present sablefish stock assessment.

2. Materials and methods

2.1. Features of the raw data

Alaska sablefish age reading is conducted routinely at the Age and Growth Program of the Alaska Fisheries Science Center. The majority of the ages are usually determined by one reader, with a subsample read independently by a senior reader. For this data set, each otolith was read independently by three readers. The frequencies of the true ages of the 172 otoliths in the dataset were:

Age	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Number	1	0	4	13	28	24	23	25	18	9	10	2	4	5	4	0	1	1

The data are concentrated at ages 5–8 because the recoveries come mostly from the commercial fishery, and fish are not fully vulnerable to the fishery until age 6. Thereafter they are subjected to both fishing and natural mortality, so the number of recoveries eventually declines with age as the number of survivors declines.

The distribution of assigned ages about true ages is quite similar for the three readers (Fig. 1). Among-reader differences are not of interest in this paper because our purpose is to characterize the distribution of age readings as a whole so that a single age misclassification matrix can be constructed for the stock assessment. If there were large differences among the readers of the otoliths in our dataset that would of course be a problem because the results would be strongly influenced by the makeup of the small sample of readers (3) that provided the data. But here the readers are similar, and we shall treat the three readings of each otolith as replicate readings by a composite reader. In effect, we combine the among-reader and within-reader variances into a single variance among replicate readings. If the among-reader differences were of primary importance, and not construction of a misclassification matrix, a mixed-effect model approach could be useful by treating otoliths as random effects (Davidian and Giltinan, 2003; Candy et al., 2012).

The age readings are biased high by about a year in some of the younger age groups (Fig. 2). Thereafter, the bias becomes steadily more negative with age. To account for the correlation among replicate readings of the same otolith, the standard deviation of the mean shown in Fig. 2 was computed by bootstrapping the data, i.e. resampling the otoliths from each age group with replacement to generate many samples and then computing the standard deviation of the bootstrap means.

When the individual errors (assigned age minus true age) are plotted otolith by otolith (Fig. 3), it can be seen that the replicate readings for a single otolith are often all high or all low, suggesting that there is some variation among otoliths of a given true age in the

Table 1
Variance components of age readings at true age.

True age	Sample size	Total variance	Due to readings	Due to otoliths	Proportion due to otoliths
4	13	1.62	1.31	0.32	0.20
5	28	3.00	1.13	1.87	0.62
6	24	1.74	0.83	0.90	0.52
7	23	1.89	1.10	0.79	0.41
8	25	6.41	2.77	3.63	0.57
9	18	3.21	1.44	1.76	0.55
10	9	4.64	3.15	1.49	0.32
11	10	7.03	3.10	3.93	0.56

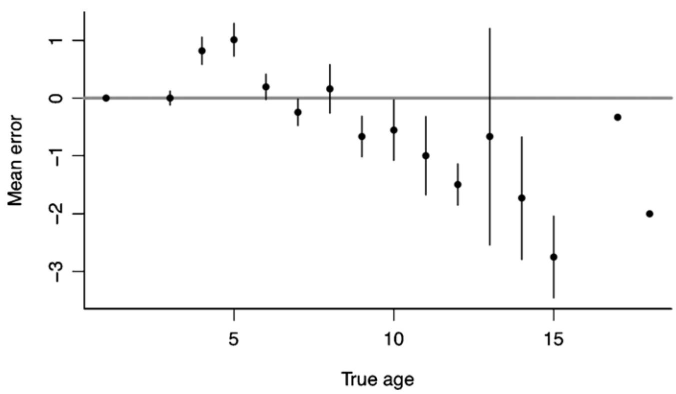

Fig. 2. Mean error (assigned age − true age) at each age ± one standard deviation of the mean. The standard deviations were computed by bootstrapping the data, so ages with a single otolith have an estimated standard deviation of zero.

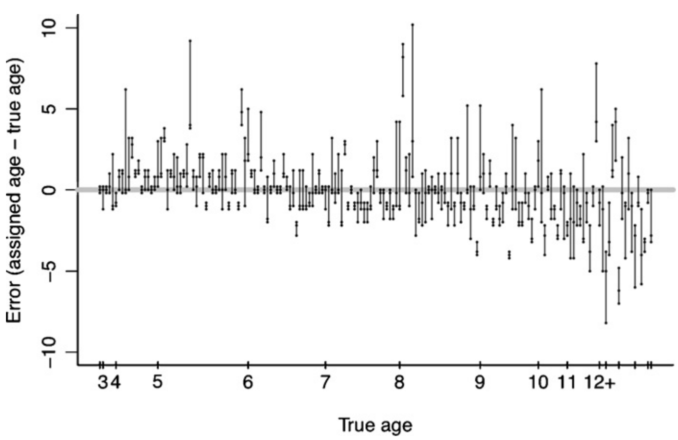

Fig. 3. Age reading errors plotted for each otolith (three points joined by a vertical line). The otoliths are sorted by age; the beginning of each age group is shown on the horizontal axis.

modal age assigned by age readers. The total variance of assigned age at a given true age can be represented as a sum of the variance among replicate readings of each otolith and the variance in modal age among otoliths, or $\sigma^2_{\text{total}} = \sigma^2_{\text{readings}} + \sigma^2_{\text{otoliths}}$. For all the ages with a useful sample size (9 or more otoliths), the two variance components are of similar magnitude (Table 1).

The empirical distribution of the errors at each age is similar over the age range 5–8, where most of the data lie, so the data for these ages can be pooled to investigate the form of the distribution (Fig. 4). The distribution is positively skewed. Negative values (underages) are more common than positive ones, but the positive limb extends farther. This error distribution reflects both the variation among otoliths in modal assigned age and the variation among replicate readings of each otolith, and is therefore a mixture of distributions rather than the distribution of readings of any one

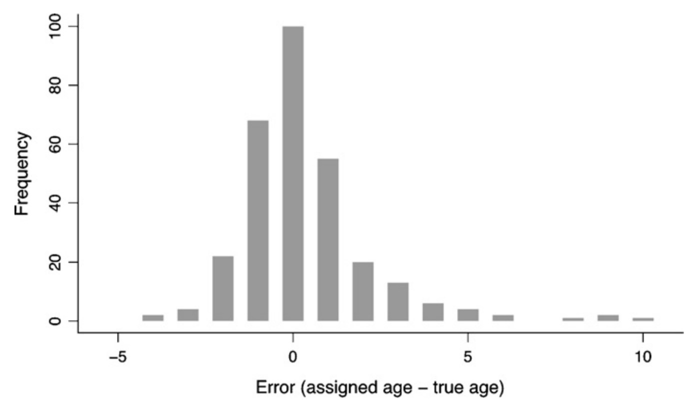

Fig. 4. Distribution of age reading errors for sablefish of ages 5–8 pooled. There are three readings of each otolith in the counts.

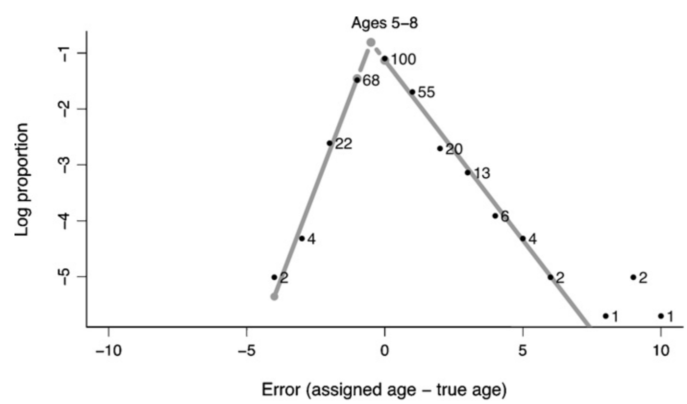

Fig. 5. Log frequencies of age reading errors plotted against error sizes, for all fish of true ages 5–8. The gray lines are fitted geometric distributions. The numbers shown at each point are the actual numbers of readings.

otolith. Our aim is to characterize this distribution mixture for the purpose of constructing an age misclassification matrix.

2.2. Parametric form of the error distribution

Clark (2004) found that age reading errors for Pacific halibut (*Hippoglossus stenolepis*) followed a geometric distribution, meaning that the probability of an error (assigned age minus true age) of size x was $P(|x|) \propto p \cdot q^{|x|}$ for $|x| = 0, 1, 2, \ldots$ for $|x|=0, 1, 2, \ldots$ where $q = 1 - p$. On a logarithmic scale, the geometric distribution is $\log P(|x|) = \text{constant} + \log p + |x| \cdot \log q$, meaning that the log of the frequency of the errors varies linearly with the size of the errors. To see whether the sablefish errors are also geometric, we plot the log frequencies of the errors for pooled ages 5–8 against the error sizes (Fig. 5). As in the case of Fig. 4, we choose this age range for an exploratory plot because it contains the bulk of the data and the empirical error distributions for the separate ages, while noisy, are similar.

Except for a very small number of large positive errors, the log frequencies of the sablefish errors for ages 5–8 pooled do in fact vary linearly with the error size (Fig. 5), so the distribution is a kind of geometric. In fact it consists of a geometric on each side of an apex which could be thought of as an unobserved (because non-integer) mode. The distribution is not symmetric, however, so the right and left hand sides have different geometric parameters, and there is a third parameter that specifies the proportions of the total distribution lying on each side, which assures that the proportions sum to one. Specifically, we have geometric parameters p_r for the right side, p_l for the left side, and s_r for the share of the total distribution falling on the right side. Let X_r denote the first (smallest, leftmost) error value on the right hand side and $X_l = X_r - 1$ the last (largest, rightmost) value on the left side. Then the probability of an error $x \geq X_r$ on the right side is $P(x) = s_r \cdot p_r \cdot q_r^{|x-X_r|}$. Similarly, the probability of an error $x \leq X_l$ on the left side is $P(x) = (1 - s_r) \cdot p_l \cdot q_l^{|x-X_r|}$. Technically, X_r, the dividing point between right and left sides, could also be considered a parameter. With smaller data sets the choice is not always clear and two or more values of X_r may need to be tried to see which results in a greater likelihood. That was not an issue with any of the fits reported here.

To investigate variations in the parameter estimates with age in the sablefish data, the asymmetrical geometric model was fitted to each age group that had at least 50 readings, as all the modal ages did. Age groups were pooled among young and old fish to reach 50 readings. To remove the influence of the few large errors that cannot be fitted, all errors larger than 6*y* were left out of the fits. Only 7 of the 516 age readings had errors larger than 6*y*.

2.3. Error distributions estimated with naive methods

Known-age data are extremely rare, so typically error distributions are estimated by methods that we term naïve because they assume no bias and fit an assumed parametric distribution to the age readings. The candidate distributions are the discrete analog to the normal distribution used by Richards et al. (1992) and Punt et al. (2008) and the symmetrical geometric that Clark (2004) found to describe the distribution of Pacific halibut (*H. stenolepis*) age readings. The continuation ratio method of Candy et al. (2012) is not used in the simulated comparisons, but we apply it for the stock assessment trials using the sablefish reader agreement data described below. Below we compare the naïve estimates of error distributions at each age to the true (asymmetrical geometric) error distributions. We also compare the performance of the naïve estimates and true error distributions when predicting the distribution of assigned ages from a true sablefish age composition containing many year-classes of varying strength, as is done in the sablefish assessment.

2.4. Application to the Alaska sablefish stock assessment

The Alaska sablefish stock is assessed annually by fitting an age-structured model to survey and commercial catch rates (which are treated as indices of abundance), commercial and survey catch age compositions where age data are available, and survey and commercial length compositions where age data are not available (Hanselman et al., 2011). Catch rate data are treated as lognormal and catch composition data as multinomial. Natural mortality and growth schedule (mean and variance of length at age) are estimated externally. The model fit includes estimates of present and historical abundance (including annual recruitments), fishing mortalities, selectivities, and catchabilities.

The present sablefish assessment model includes an age misclassification matrix based on the aging error estimates obtained by Heifetz et al. (1999), who fitted normal distributions to their known-age data, thus estimating both bias and variance at each age. Given the assessment model's internal estimates the true age composition of fishery or survey catches, this matrix is used to predict the observed distribution of reader-assigned ages. If known-age fish were unavailable for the assessment, an age misclassification matrix would have been used that estimated the error from the large number of replicate readings that have been done for quality control for many years. For this study, we calculated aging error using the methods of Kimura and Lyons (1991) and used the approaches of Richards et al. (1992), Candy et al. (2012), and the present study to construct naïve misclassification matrices.

We investigated the effect on the Alaska sablefish assessment of five treatments of aging error: (i) treating the assigned ages as true (i.e., not using any misclassification matrix), (ii) using the naïve normal misclassification matrix, and (iii) using a naïve continuation-ratio misclassification matrix of Candy et al. (2012), (iv) using a naïve misclassification matrix based on the geometric model and (v) using a misclassification matrix based on the geometric model with true ages described in section 2. For the continuation-ratio misclassification matrix, parameters were estimated for seven groups of ages (1–2, 3–5, 6–8, 9–11, 12–14, 15–17, and 18+, similar to the grouping in Candy et al. (2012) and in geometric comparisons later). For the naïve geometric misclassification matrix, s_r was set at 0.5 (symmetric) and p_r and p_l were estimated for ages 2–31 by least squares fit to the observed reader agreement data. In the known-age geometric misclassification matrix, all age 1 fish are assumed to be aged correctly, consistent with the one otolith in the data. Ages 3–5 have an error distribution generated with the parameters shown in the lower part of Table 2. The assumed error distribution at age 2 (where there are no data) is the simple average of the age 1 and age 3–5 distributions. Beginning at age 6, the geometric parameters of the error distribution at each age are calculated from the fitted asymptotic functions (Fig. 6). For each true age, the asymmetric distribution was computed (where feasible) out to errors of $\pm 10y$, where the probabilities are vanishingly small at all ages, and then rescaled to sum to one.

3. Results

3.1. Parameter estimates of the geometric error distribution

Maximum-likelihood estimates of the parameters of the asymmetric geometric distribution for each age group were located numerically, treating the error frequencies as multinomial and calculating the multinomial proportions for all errors using the three geometric parameters p_l, p_r, and s_r as explained above. Variances of the parameter estimates were computed by bootstrapping, i.e. sampling the otoliths from each age group with replacement and refitting the model for each bootstrap sample. The parameter estimates are shown in the first part of Table 2. Through age 5 there are no negative errors beyond -1, so the estimate of the left geometric parameter p_l is 1. The value of p_l decreases steadily with age, meaning that the spread of the negative errors grows with age. The right geometric parameter p_r varies much less with age.

The parameter estimates are similar for age groups 3–4 and 5, for ages 6 and 7, for ages 8 and 9, and for age groups 10–11 and 12–18. Parameter estimates for the combined groups are shown in the second part of Table 2 and the fits are plotted in Fig. 7. Plots of these fits to the data in each group show that the geometric distribution fits the form of the empirical distribution well across the range of ages in the data.

From age 6 onward, the estimated geometric parameters decrease smoothly with age on a course that is well

Table 2
Parameter estimates from fits of the asymmetric geometric model to subsets of the data grouped by true age. In all fits the dividing point between right and left limbs was $X_r = 0$. Errors larger than 6 years were excluded from the fits. The last three columns are the parameter estimates ± one standard deviation.

Ages	Number of readings included (excluded)	p_l	p_r	s_r
Group data to obtain at least 50 readings per group				
3–4	51 (0)	1	0.55 ± 0.06	0.90 ± 0.04
5	83 (1)	1	0.44 ± 0.05	0.84 ± 0.05
6	72 (0)	0.74 ± 0.12	0.60 ± 0.08	0.76 ± 0.06
7	69 (0)	0.77 ± 0.05	0.56 ± 0.07	0.51 ± 0.07
8	72 (3)	0.62 ± 0.08	0.52 ± 0.10	0.56 ± 0.07
9	54 (0)	0.54 ± 0.08	0.52 ± 0.06	0.43 ± 0.10
10–11	56 (1)	0.44 ± 0.04	0.47 ± 0.10	0.39 ± 0.08
12–18	49 (2)	0.35 ± 0.04	0.41 ± 0.11	0.35 ± 0.08
Combine groups by similar parameter estimates				
3–5	134 (1)	1	0.47 ± 0.04	0.87 ± 0.04
6–7	141 (0)	0.76 ± 0.05	0.58 ± 0.05	0.64 ± 0.06
8–9	126 (3)	0.58 ± 0.06	0.52 ± 0.07	0.50 ± 0.06
10–18	105 (3)	0.38 ± 0.03	0.44 ± 0.07	0.37 ± 0.06

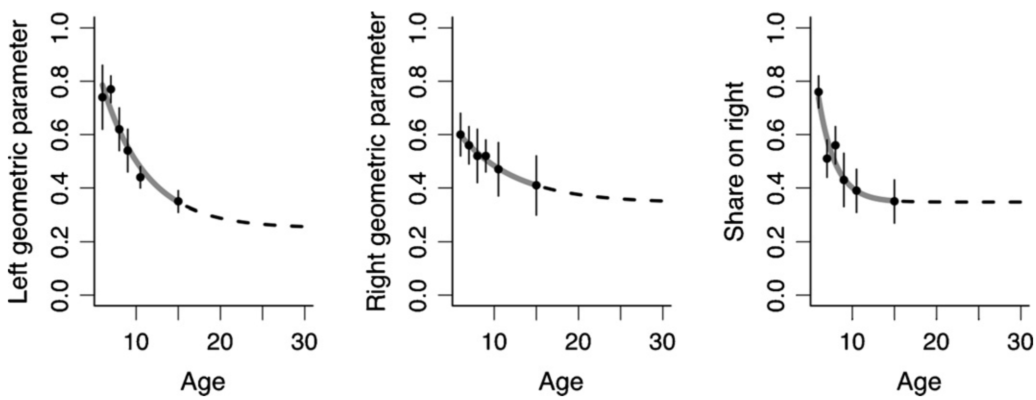

Fig. 6. Variation of geometric parameter values with age (ages 6–9, 10–11, 12–18). The points are parameter estimates by age group, the gray line is a fitted asymptotic function, and the broken line is the extrapolation of the function to older ages where there are no data.

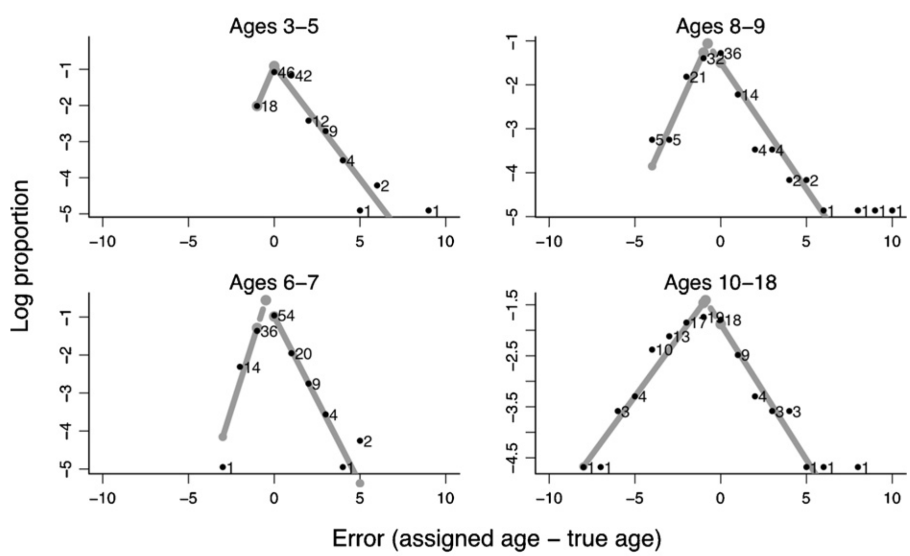

Fig. 7. Log frequencies of age reading errors plotted against error sizes for the entire data set, grouped by age according to similarity of parameter estimates (Table 2). The gray lines are fitted geometric distributions. The numbers shown at each point are the actual numbers of readings.

approximated by an asymptotic function of age (Fig. 6), meaning that the parameter values appear to level out among older age groups. Lacking any known-age data for the older age groups, those functions can be extrapolated to estimate the distribution of age readings of older fish. This sort of extrapolation is necessary to construct the misclassification matrix for the assessment, which goes out to age 31. The parameters were estimated numerically by ordinary least squares so as to achieve a good fit across the entire range of ages rather than favoring the ages with the most data. Extra weight was placed on the last data point to assure good agreement with the data at the beginning of the extrapolation.

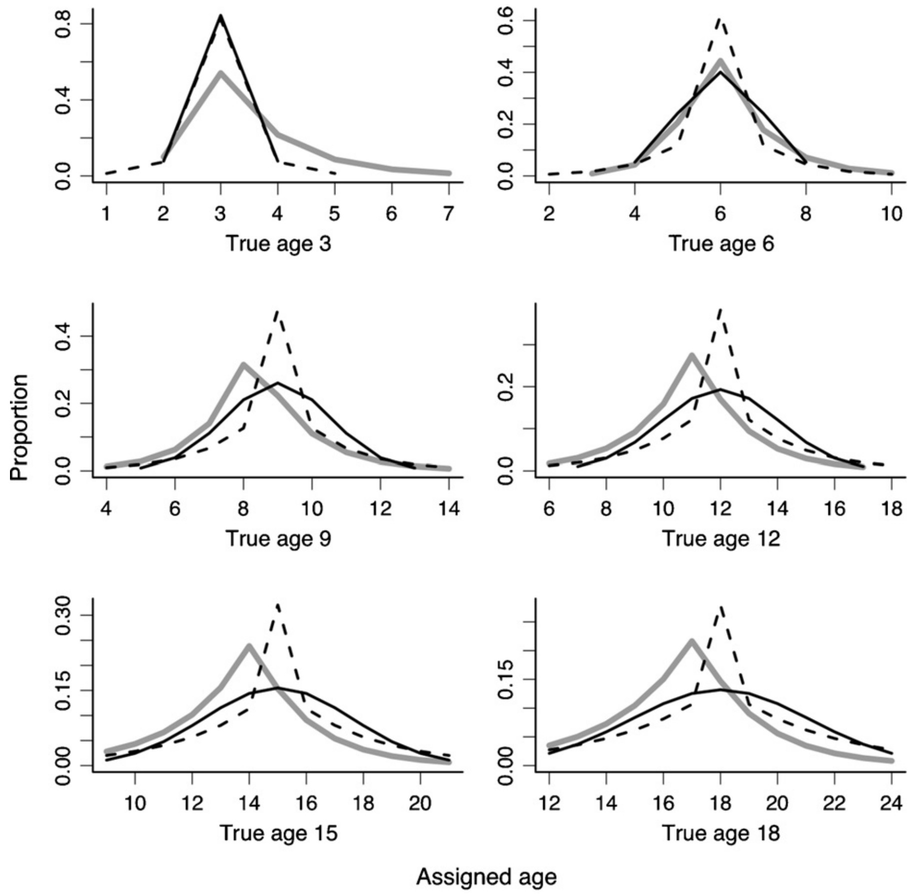

Fig. 8. True error distributions (gray lines) and estimates obtained by fitting the naïve normal model (solid lines) and naïve geometric model (broken lines) without benefit of the known ages. The true distributions are asymmetric geometrics; the fitted distributions are symmetric.

3.2. Comparison to naïve methods

Fig. 8 compares the predicted distribution of age readings at selected true ages calculated using the asymmetrical geometric model based on known-age data with the predictions calculated using the naïve normal and geometric models based only on the replicate readings. Comparisons of the distributions are shown for true ages 3, 6, 9, 12, 15, and 18. One would expect these methods to provide poor estimates when applied to the sablefish data (without the known ages) because they assume the wrong form of distribution, they assume the readings are unbiased, and they cannot detect the variance added by differences among otoliths. The naïve predictions are indeed wrong, but not by as much as one might expect. The naïve geometric estimates are too peaked in the center because they underestimate the true variance. The naïve normal fits also underestimate the variance but because of their more diffuse form they nevertheless approximate the shape and spread of the true distributions quite well over a wide range of ages. They are of course increasingly shifted to the right of the true distributions by the bias present in the latter at higher ages. If there were no bias, the normal fits would appear to be quite serviceable for most of the ages plotted.

What matters for stock assessment purposes is not the closeness of the estimated error distributions at each age but how well or poorly the estimates predict the actual distribution of age readings that result from a typical true age distribution containing the full range of ages. This is the prediction made within a stock assessment by a misclassification matrix in order to fit the assessment model to the age reading data. Fig. 9 shows a set of predictions for one random draw of the year-class strengths present in a sablefish-like stock (fishing and natural mortality both 0.1, full fishery selection at age 6, lognormal recruitment variability with standard deviation 1). The points are the true proportions at age in the catch and the gray line is the expected distribution of age readings that results from the asymmetrical geometric distributions estimated from the known-age data above. The other lines show the predictions made by misclassification matrices based on the naive normal and naïve geometric model fits.

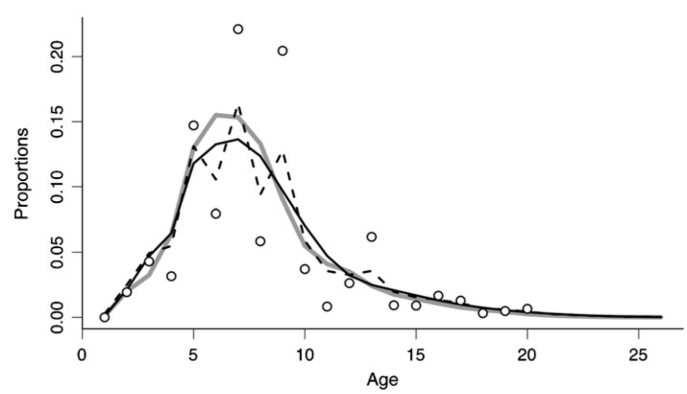

Fig. 9. Actual and predicted sablefish age reading distributions in a sample case. Points are true proportions at age in the catch, randomly generated with recruitment variability and other parameter values appropriate for sablefish. Solid gray line is the predicted distribution of readings with the true error distribution (based on known ages), solid black line is the prediction from the fitted naive normal model, broken line is the prediction from the fitted naïve geometric model.

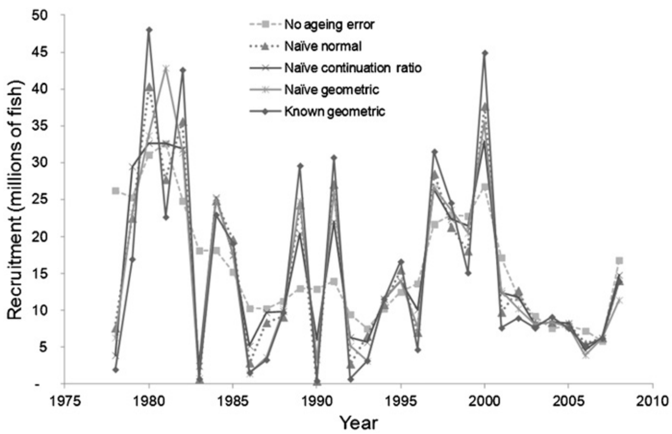

Fig. 10. Estimated sablefish recruitment with five ageing error treatments from 1978 to 2008. The naïve methods are estimated from reader agreement data, while the known geometric are from known-age sablefish.

The observed age reading distribution is quite different from the true age distribution for all three misclassification matrices, because they all predict that the variance of age readings will tend to assign many fish from more numerous to less numerous neighboring age groups, thereby effectively smoothing the age distribution. What is remarkable is how similar the three predictions are in this case. Evidently the naive estimates, even though they do not match the true error distributions very well, go a long way toward predicting the actual pattern of smoothing that takes place.

The degree of concordance between the true distribution of age readings and the naïve predictions varies considerably among random draws of year-class strengths. Sometimes they are very close and sometimes quite different. The example shown in Fig. 9 is perhaps a bit closer than average but not exceptional. It also shows a common feature of most examples; the naïve geometric prediction tends to be more variable than the naïve normal.

3.3. *Effect of different error treatments on the sablefish stock assessment*

The five treatments produce almost the same estimate of mean recruitment and therefore current estimates of current abundance (Fig. 10). When aging error is ignored, recruitment variability is greatly underestimated. The geometric error distribution based on known ages results in the highest recruitment variability of the five methods; the large year classes are estimated to be larger and the weak year classes are estimated to be weaker. These results agree with Punt et al. (2008) who showed for blue grenadier that assuming age readings were exact led to the sizes of weak cohorts being overestimated and those of strong cohorts being underestimated. The inclusion of bias in the geometric aging error matrix based on known ages has little effect on the temporal location of year classes when compared to the naïve normal model. The no aging error, naïve continuation-ratio and naïve geometric methods resulted in some differences on the size and location of several year classes in the early 1980s compared to the known-age estimates (Fig. 10).

Another effect of ignoring aging error was that the precision of the model estimates of year-class strengths was overstated. Accounting for aging error showed that only the strong year classes were estimated with reasonable precision. For example, the estimates of the strong 1997 and 2000 year classes had a coefficient of variation (CV) of 19 and 20% respectively using the known-age geometric model and 10% and 11% when aging error was ignored. For the weak 1990–1996 year classes the geometric model showed CVs of 16–358% while ignoring aging errors produced CVs of 13–16%.

4. Discussion

Our study utilized a unique data set among marine species to explore true bias and variability of otolith derived ages. Age validation is essential for sustainable management, and the use of known-age fish from mark-recapture studies is one of the few ways to obtain this information, particularly from wild fish (Campana, 2005). We found that sablefish age readings are somewhat biased and that bias changes from positive to negative as fish get older. In common with Beamish and McFarlane (2000), we found that ages of older sablefish were consistently underestimated. The aging error variability was well-described with a geometric distribution, which was readily translated into a misclassification matrix for use in stock assessment.

Accounting for age reading error in stock assessments is desirable in order to obtain accurate estimates of year-class strengths and variances of the estimates, particularly for recent year-classes in the stock that are used as the basis for stock projections and setting annual catch limits. Ideally estimates of age reading error would be made with data from known-age fish, as in the case reported here, but such data are seldom available, so error estimates are usually made with naïve methods such as the naïve normal that assumes unbiased readings of every otolith and a normal distribution whose variance is estimated from replicate readings. Our results suggest that the naïve estimates can actually perform quite well despite being based on false assumptions, and agree with Punt et al. (2008) that some estimate of aging error should be used routinely, rather than ignoring aging error altogether. The naïve estimates can be expected to underestimate the variance of age readings, but the underestimates perform much better than an estimate of zero error in the sablefish assessment and nearly as well as the true error distribution based on known ages. The most commonly employed misclassification matrix, the naïve normal, had slightly different results from the geometric distribution using known ages, but in terms of determining year class strength and location, it performed the best of the naïve methods in this case. This is a comforting result given the lack of known-age data to construct misclassification matrices for most assessments. It is of course possible that the naive methods would fail badly when applied to readings with a larger bias.

Acknowledgments

We thank Steve Candy for providing assistance with implementing the continuation-ratio method. We also thank Andre Punt and an anonymous reviewer for careful reviews that greatly improved the paper. Pete Hulson and Tom Helser provided helpful reviews of an early draft. The findings and conclusions in this paper are those of the authors and do not necessarily represent the views of the National Marine Fisheries Service.

References

Beamish, R.J., McFarlane, G.A., 2000. Reevaluation of the interpretation of annuli from otoliths of a long-lived fish, *Anoplopoma fimbria*. Fish Res. 46, 105–111.

Campana, S.E., 2005. Otolith science entering the 21st century. Mar. Freshw. Res. 56, 485–495.

Candy, S.G., Nowara, G.B., Welsford, D.C., McKinlay, J.P., 2012. Estimating an ageing error matrix for Patagonian toothfish (*Dissostichus eleginoides*) otoliths using between-reader integer errors, readability scores, and continuation ratio models. Fish. Res. 116, 14–23.

Cardinale, M.P., Doering-Arjes, P., Kastowsky, M., Mosegaard, H., 2004. Effects of sex, stock, and environment on the shape of known-age Atlantic cod (*Gadus morhua*) otoliths. Can. J. Fish. Aquat. Sci. 61, 158–167.

Clark, W.G., 2004. Nonparametric estimates of age misclassification from paired readings. Can. J. Fish. Aquat. Sci. 61, 1881–1889.

Clark, W.G., Hare, S.R., 2006. Assessment and management of Pacific halibut: data, methods, and policy. Int. Pac. Halibut Comm. Sci. Rep., 83.

Conn, P.B., Diefenbach, D.R., 2007. Adjusting age and stage distributions for misclassification errors. Ecology 88, 1977–1983.

Davidian, M., Giltinan, D., 2003. Nonlinear models for repeated measurements: an overview and update. J. Agric. Biol. Environ. Stat. 8, 387–419.
Hanselman, D.H., Lunsford, C.R., Rodgveller, C.J., 2011. Assessment of the sablefish stock in Alaska. In: Stock Assessment and Fishery Evaluation Report for the Groundfish Resources of the Gulf of Alaska, North Pacific Fishery Management Council, 605W. 4th Avenue, Suite 306, Anchorage, AK 99501, pp. 329–468.
Heifetz, J., Anderl, D., Maloney, N.E., Rutecki, T.L., 1999. Age validation and analysis of ageing error from marked and recaptured sablefish, *Anoplopoma fimbria*. Fish. Bull. 97, 256–263.
Kimura, D.K., Lyons, J.J., 1991. Between reader bias and variability in the age-determination process. Fish. Bull. 89, 53–60.
McBride, R.S., Hendricks, M.L., Olney, J.E., 2005. Testing the validity of Cating's (1953) method for age determination of American shad using scales. Fisheries 30 (10), 10–18.
McDougall, A., 2004. Assessing the use of sectioned otoliths and other methods to determine the age of the centropomid fish, barramundi (*Lates calcarifer*) (Bloch), using known-age fish. Fish. Res. 67, 129–141.
Pearson, D.E., Shaw, F.R., 2004. Sources of age determination errors for sablefish (*Anoplopoma fimbria*). Fish. Bull. 102, 127–141.
Punt, A.E., Smith, D.C., KrusicGolub, K., Robertson, S., 2008. Quantifying age-reading error for use in fisheries stock assessments, with application to species in Australia's southern and eastern scalefish and shark fisheries. Can. J. Fish. Aquat. Sci. 65, 1991–2005.
Richards, L.J., Schnute, J.T., Kronlund, A.R., Beamish, R.J., 1992. Statistical models for the analysis of ageing error. Can. J. Fish. Aquat. Sci. 49, 1801–1815.